QF 604
ECONOMETRICS OF FINANCIAL MARKETS

LECTURE 10

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LECTURE OUTLINE

01 FRACTIONAL INTEGRATION

02 ARFIMA

03 LONG MEMORY VOLATILITY MODELS

04 TESTING FOR FI

00 Presence of Long Term Memory in Time Series

- An empirical fact is the persistence of autocorrelations in financial time series. This persistence after many lags is called long-term memory.
- The extent of the persistence is consistent with a stationary process. But autocorrelations can take much longer to decay than the exponential rate associated with the ARMA class.
- The autocorrelation function exhibits persistence that is neither consistent with an I(1) process nor an I(0) process.
- The fractionally differenced I(d) (where d is a fractional number) process can be regarded as a halfway house between the I(0) and I(1) paradigms.
- One plausible advantage of long memory models is that they imply different long run predictions and effects of shocks than in conventional macroeconomics approaches.

01 Fractional Integration

Consider the model

$$\Delta^d y_t = (1 - L)^d y_t = u_t \tag{2.1}$$

where u_t is an independent identically distributed (*iid*) disturbance with $\mathrm{E}(u_t) = 0$ and $\mathrm{var}(u_t) = \sigma^2$. When $d = 0$, then y_t is simply white noise and is *weakly stationary*. When $d = 1$ then y_t is a random walk process and is *non-stationary*. In this case we say that y_t is integrated of order 1, and write $y_t \sim I(1)$. We now want to consider the properties of (2.1) for *fractional* values of d lying between 0 and 1.

We require the generalized binomial expansion to substantiate the concept of lagged operators above. Lagged operators follow algebraic properties of a deterministic variable. For the generalized binomial expansion, we require understanding of the Taylor series expansion.

The Taylor series expansion of f(x), provided it converges, is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots$$

Maclaurin series and its polynomials are Taylor polynomials $f^{(n)}(a)$ centered at zero, i.e., $f^{(n)}(0)$.

01 Fractional Integration

Generalized Binomial Theorem

Generalized Binomial Theorem. The Generalized Binomial Theorem allows us to express $(1+x)^r$ as a Maclaurin series using a natural generalization of the binomial coefficients. For any $r \in \mathbb{R}$ and $n \in \mathbb{N}_0$, we set

$$\binom{r}{n} := \frac{r(r-1)\cdots r-n+1}{n!}. \tag{0.3}$$

Observe that when $r \in \mathbb{N}_0$, we recover the standard formula for the binomial coefficients. We are now in a position to generalize the Binomial Theorem.

Theorem 1. *For any $r \in \mathbb{R}$,*

$$(1+x)^r = \sum_{n=0}^{\infty} \binom{r}{n} x^n. \tag{0.4}$$

Proof. Set $f(x) = (1+x)^r$. For each $n \in \mathbb{N}_0$, we see that $f^{(n)}(x) = (r)_n(1+x)^{r-n}$, and so $f^{(n)}(0)/n! = \binom{r}{n}$. Therefore the Maclaurin formula of $f(x)$ is that one in the right-hand side of (0.4). $\square$

Since Maclaurin or Taylor series apply for any r $\in$ **R**, it applies to the generalized binomial coefficients in the same expansion.

01 Fractional Integration

Binomial Theorem

$$\Delta^d = (1-L)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-L)^k \tag{2.2}$$

where the binomial coefficients $\binom{d}{k}$ are defined by

$$\binom{d}{k} = \frac{d(d-1)(d-2)\cdots(d-k+1)}{k!}$$

and $k!$ (read as k *factorial* or k *shriek*) is

$$k! = 1 \cdot 2 \cdot 3 \cdot \cdots \cdot (k-1) \cdot k$$

with $0! = 1$.

Using these definitions, we can rewrite (2.2) as

$$(1-L)^d = 1 - dL + \frac{d(d-1)}{2!}L^2 - \frac{d(d-1)(d-2)}{3!}L^3 + \cdots . \tag{2.3}$$

Provided that -0.5 < d < 0.5, the y_t process is weakly stationary and invertible. This range of d is the most relevant. Also, persistent autocorrelation occurs for d > 0.

01 Fractional Integration

For integral d, the coefficients will be zero for $k > d$ and so the infinite series is truncated at order d. Thus, for example, we have

$$(1-L)^2 = 1 - 2L + L^2$$

For fractional d, $0 < d < 1$, the infinite series will not be truncated. For example

$$(1-L)^{0.5} = 1 - 0.5L + \frac{-0.25}{2}L^2 - \frac{.375}{6}L^3 + \frac{-0.9375}{24}L^4 - \frac{3.28125}{120}L^5 + \cdots$$

which is an infinite lag expansion. Depending on the value of d, the fractionally differenced series $(1-L)^d y_t$ may be stationary or non-stationary. It turns out that the condition for stationarity is that $d < 0.5$. Thus for the example case $d = 0.5$, the sum of the infinite series of coefficients, 1, -0.5, -0.125, -0.0625, -0.0390625, -0.02734375 , $\cdots$, does not converge fast enough.

01 Fractional Integration

Stationary Fractionally Integrated Processes

It can be shown that, for $d < 0.5$, the process (2.1) is *weakly stationary* and has an *infinite order MA (Wold)* representation given by

$$y_t = (1 - L)^{-d} u_t = \sum_{i=0}^{\infty} \theta_i u_{t-i} \tag{2.4}$$

where

$$\theta_k = \frac{d(1+d)(2+d)\cdots(k-1+d)}{k!}.$$

For proof, see C. W. J. Granger & Roselyne Joyeux, 1980. "An Introduction To Long-Memory Time Series Models And Fractional Differencing," Journal of Time Series Analysis, Wiley Blackwell, vol. 1(1), pages 15-29, January.

01 Fractional Integration

Stationary Fractionally Integrated Processes

A comparison between the autocorrelation function of a stationary fractionally integrated model and that of a simple *short-memory* $AR(1)$ process can be illustrated by an example.

Comparison of ACF of fractionally integrated (FI) and AR(1) model.
Source: Campbell, Lo and MacKinlay (1997)

Lag	*FI* ($d = \frac{1}{3}$)	*AR(1)* $\phi = 0.5$
1	0.500	0.500
2	0.400	0.250
3	0.350	0.125
4	0.318	0.063
5	0.295	0.031
10	0.235	0.001
25	0.173	2.98×10^{-8}
50	0.137	8.88×10^{-16}
100	0.109	7.89×10^{-31}

The table makes clear the big difference between the *geometric decay* of the ACF coefficients in the $AR(1)$ model and the *hyperbolic decay* of the (stationary) *fractionally integrated* (FI) model. After *100* periods, the FI model still exhibits large autocorrelation while in the $AR(1)$ model it is completely negligible.

01 ARFIMA MODELS

ARIMA models can be generalised to incorporate fractional integration, leading to the *ARFIMA(p,d,q)* model:

$$\phi(L)\Delta^d y_t = \theta(L)u_t \tag{3.1}$$

where

$$\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p$$

and

$$\theta(L) = 1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta_q L^q$$

and where d now represents the order of fractional differencing.

The model can be rewritten as

$$\Delta y_t = \Delta^{1-d}\phi(L)^{-1}\theta(L)u_t = A(L)u_t$$

A more satisfactory approach is to estimate d as part of the joint maximum likelihood estimation of the parameters of the *ARFIMA* model. The log-likelihood of the *ARFIMA* model is given by

$$\log L(\mathbf{y}; d, \boldsymbol{\phi}, \boldsymbol{\theta}, \boldsymbol{\Sigma}) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\log|\boldsymbol{\Sigma}| - \frac{1}{2}\mathbf{y}'\boldsymbol{\Sigma}^{-1}\mathbf{y}$$

where

$$\boldsymbol{\Sigma} = \mathrm{var}(\mathbf{y})$$

is the variance covariance matrix of $\mathbf{y}$ which is a (complicated) function of the unknown model parameters d, $\boldsymbol{\phi}$ and $\boldsymbol{\theta}$.

This approach is developed by Sowell (1992) and is adopted in the estimation package *PcGive* by Doornik and Hendry (2001).

Assuming u_t is normally distributed

01 LONG MEMORY VOLATILITY MODELS

Long memory in the conditional variance of financial data series has led to the development of fractionally integrated versions of both the GARCH and the stochastic volatility model.

GARCH (1,1) $\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2$

can be rewritten as the *ARMA* process

$$u_t^2 = \alpha_0 + (\alpha_1 + \beta_1) u_{t-1}^2 + \varepsilon_t - \beta_1 \varepsilon_{t-1}$$

$$u_t^2 = \alpha_0 + (\alpha_1 + \beta_1) u_{t-1}^2 + (u_t^2 - \beta_1 u_{t-1}^2) - (\sigma_t^2 - \beta_1 \sigma_{t-1}^2)$$

where $\varepsilon_t = u_t^2 - \sigma_t^2$. For the case of the *FIGARCH(1,d,1)* model, by analogy, we have

$$\Delta^d u_t^2 = \alpha_0 + (\alpha_1 + \beta_1) \Delta^d u_{t-1}^2 + \varepsilon_t - \beta_1 \varepsilon_{t-1},$$

which can be rewritten as

$$\sigma_t^2 = \alpha_0 + (1 - \Delta^d) u_t^2 - (\beta_1 - (\alpha_1 + \beta_1)\Delta^d) u_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \qquad (4.1)$$

01 LONG MEMORY VOLATILITY MODELS

Long Memory Stochastic Volatility Model

Breidt, Crato and de Lima (1998) develop a simple stochastic volatility model with long memory. The model is defined by

$$y_t = \sigma_t u_t \tag{4.2}$$

where the volatility, σ_t, follows the process

$$\sigma_t^2 = \phi_1^2 \exp(\varepsilon_t) \tag{4.3}$$

and

$$(1 - L)^d \varepsilon_t = \eta_t \tag{4.4}$$

where u_t and η_t are independent, normally distributed white noise processes with unit variance and $0 < d < 0.5$.

01 LONG MEMORY VOLATILITY MODELS

Long Memory Stochastic Volatility Model

$$y_t = \sigma_t u_t \tag{4.2}$$

Squaring (4.2) and taking logarithms,

$$\begin{aligned} \log(y_t^2) &= \log \sigma_t^2 + \log u_t^2 \\ &= \log \phi_1^2 + \varepsilon_t + \log u_t^2 \\ &= (\log \phi_1^2 + \mathrm{E}(\log u_t^2)) + \varepsilon_t + (\log u_t^2 - \mathrm{E}(\log u_t^2)) \end{aligned}$$

or

$$\log(y_t^2) = \mu + \varepsilon_t + e_t \tag{4.5}$$

where $e_t = \log u_t^2 - \mathrm{E}(\log u_t^2)$ is a mean-zero disturbance term which is independent of η_t but which has a non-normal distribution. Equations (4.5) and (4.4) represent a model which can be estimated by quasi-maximum likelihood. Estimating this model on squared daily returns on companies in the S&P 500 index, Bollerslev and Jubinski (1999) found a median estimate of d of about 0.38.

01 TESTING FOR FRACTIONAL INTEGRATION

The Range over Standard Deviation R/S (also called the rescaled range) Test

A test for fractional integration was originally proposed in the engineering literature by Hurst (1951) and applied to financial economics by Mandelbrot (1972).

This is called the '*range over standard deviation*' or R/S statistic and is defined by

$$R_0 = \widehat{\sigma}_0^{-1}\{\max_i \sum_{t=1}^{i}(y_t - \overline{y}) - \min_i \sum_{t=1}^{i}(y_t - \overline{y})\} \tag{5.1}$$

where

$$\widehat{\sigma}_0^2 = \frac{\sum_{t=1}^{T}(y_t - \overline{y})^2}{T}$$

is an estimate of the sample variance. The term in braces in (5.1) is called the range and is the difference between the largest and smallest partial sums of mean deviations of y_t. Since the sum of mean deviations over all observations is zero, this difference will always be non-negative.

The range will increase with the sample size T. A plot of log R/S vs. log sample size gives slope which is an estimate of Hurst exponent H. If H = ½, time series is a random walk.

01 TESTING FOR FRACTIONAL INTEGRATION

Lo (1991) suggested a correction to (5.1) in order to avoid short-run stationary process from influencing a larger R/S statistic:

$$R_q = \widehat{\sigma}_q^{-1}\{\max_i \sum_{t=1}^{i}(y_t - \overline{y}) - \min_i \sum_{t=1}^{i}(y_t - \overline{y})\} \qquad (5.2)$$

and

$$\widehat{\sigma}_q^2 = \widehat{\sigma}_0^2(1 + \frac{2}{T}\sum_{j=1}^{q} w_{qj}\widehat{\rho}_j)$$

where the w_{qj} are triangular weights defined by

$$w_{qj} = 1 - \frac{j}{q+1}$$

and $\widehat{\rho}_j$ is the sample autocorrelation coefficient defined by

$$\widehat{\rho}_j = \frac{\sum_{t=j+1}^{T}(y_t - \overline{y})(y_{t-j} - \overline{y})}{\sum_{t=1}^{T}(y_t - \overline{y})^2}.$$

The correction in (5.2) will pick up any short-run dependence in the series y_t so that the test will have power to distinguish fractionally integrated processes from other stationary processes. The appropriate choice of q is left to the user.

On the null hypothesis that $d = 0$, the statistic $T^{-1/2}R_q$ converges asymptotically to a well defined random variable for which critical values are tabled in Lo (1991). Lo shows that this test is consistent against all alternatives in the class

Reject H_0 when test statistic is too large.

PART 2 – TEST

90 mins. Closed Book.

Lockdown Browser.

1 x A4 reference sheet allowed.

Bring calculator.